

Project Portfolio

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YOM Platform

URL

<https://yomweb.luluone.com/>

Role

Lead Software Engineer (Technology Transformation)

Key Technologies

SpringBoot, Java, MongoDB, GraphQL, PostgreSQL, Oracle 12c, Apache NiFi, Solr, Kafka, HashiCorp Vault, Netflix Zuul Router, JPA, Docker, OpenShift (Kubernetes), Rancher, RestHeart, JasperReports, JSP, JavaScript, RabbitMQ, Jenkins, Dropwizard, Apache POI, Maxmind GeoIP 2, Mautic, Strapi, Apache Ambari, UIPath, Jiffy, Mobile Security Framework (MobSF).

Description

The Lulu International Exchange work on the domain of Remittance and Forex domain. YOM is very large FinTech service on which the Lulu International Exchange works. Currently, it is written in Java and JSP. Persisted with Oracle 12c. And as a single monolith Tomcat server.

I was hired as part of the introduction of new technologies, and migration towards API based (microservices) transformation.

Modules

YOM 2 (Yo2)

A collection of Microservices which will be replacing existing YOM Service. Utilizes modern technologies and SDLC concepts into the migration of current services.

PDD Core Service

An API service class which eliminates DAO pattern. This microservice utilizes Oracle Stored Procedures and publish them as APIs for the consumption of new services.

RedCurrant

RedCurrant is a collection of APIs which will interface with different Banks, Treasuries, and Remittance providers. All the transactions are routed through this service. 60 Banks and 20 remittance service APIs are registered in RedCurrant.

YOM Adaptor

This is a dedicated business transformer service, which will implement business logic to YOM transactions to match with RedCurrant registered API services. It is configured to automatically choose the best available service to send transactions to other internal/external services.



Alphabot Notifier

Alphabot Notifier is a dedicated microservice which handles the different notifications to customers, apps, and websites. Notification APIs includes emails, push notifications, and SMS through different providers for different countries.

Alphabot MIS

This microservice is written to generate different types of reports from a Solr indexed mirror data of the database.

Instio Platform

URL

<http://instio.co/>

Role

Principal Software Engineer (Backend)

Key Technologies

Dropwizard, Java, MongoDB, Microsoft Azure, Azure ML Studio, Azure Cognitive Services, TripAdvisor API, SendGrid, FullContact4J, Datumbox NLP, Freemarker, PrettyTime NLP, OkHttp, Nexmo/Twillio/2Factor SMS APIs.

Description

Instio is a guest experience management software-as-a-service (SaaS) platform that helps clients improve their customer experience and client retention, measuring with the Net Promoter Score (NPS). The system automatically measures analyses feedback and has close-loop tools to retain customers better and reduce churn. This system is comprised of a cluster of microservices working together to get the complete goal.

Modules

Instio Guest Experience Management (Gem)

Allow the guests to order room service, schedule a taxi, make a dinner reservation at the restaurant, request housekeeping, book a spa or request check out all using their own smartphone without installing any application

Instio Feedback Collection (Express)

Collect feedback from guests automatically from various touch-points - check-in, during stay, check-out or post-checkout.

Instio Marketing Automation (Pulse)

The app empowers Instio customers to focus on what impacts most of their business and improve the guest experience. By providing timely alerts, notifications and the ability to respond to guest feedback in real-time, Instio Pulse help to proactively engage with guests. Know what your guests are saying about your hotel around the clock and respond directly to them from your mobile device or tablet.

SnapRapid

URL

<http://snaprapid.com/home>

Role

Backend Engineer

Key Technologies

Java, MongoDB Async, Apache OpenNLP, Stanford CoreNLP, Lexical State Machine, Apache Tika, LangDetect, Docker, RabbitMQ, Solr/Lucene, MultiWordNet, MySQL, emojiJava, OpenCSV, YAML, FullContact4J.

Description

SnapRapid an enterprise SaaS system. This system is driven by client-specific dashboards for rights holders, brands and sponsors. SnapRapid is tracking and analyzing exposure for brands and celebrities through social media for our sporting and entertainment clients. It analyses 11 social media platforms: Twitter; Facebook; Instagram; YouTube; Weibo; Vine; DailyMotion; Tumblr; Google+; Reddit and Flickr, millions of images and many thousands of videos per day. Snaprapid uses deep learning and NLP technologies we help brands and celebrities, including athletes, to better understand their social media exposure and value to marketers.

Modules

PreProcessor

The module which is responsible for loading all collected data, message passing systems, entities, language detection, etc.

Text cleaner

Module to clean text collected from different social media platforms. Different text cleaning mechanisms are utilized to get the best possible data.

Concept Extractor

The module will collect key concepts in post content. Also sorts and catalogs data accordingly.

NER

This module has various NER technologies like OpenNLP, Stanford NLP, State Machine Lexical Analyser to collect all possible entities present in a post (English/German/Spanish/Chinese).

SynSet lookup

Wordnet based module which used to get synonyms for identified entities. Supported languages include English, Portuguese, Spanish, and Italian.

Upzel CRM Suite

URL

<http://upzel.io>

Role

Key Backend Developer

Key Technologies

Dropwizard, Java, Facebook Marketing SDK, Google AdWords SDK, Facebook Graph API, MongoDB, FullContact4J, OpenNLP NER, Apache POI, Solr/Lucene, Google Maps API, Mailgun

Description

This project was a CRM suite. This had company level logins with feature tiers. The companies are allowed to handle multiple AD publishers like Facebook Marketing and Google Adwords. They had facilities to publish Ad campaigns to various platforms from Upzel. The CRM suit also allowed to identify targeted customers from collected email lists. These email listed are matched against FullContact APIs to resolve usable CRM customer data.

BIZHIntegra Semantic Web

URL

<http://bizlem.io>

Role

Key Backend Developer

Key Technologies

Dropwizard, Java, Apache Stanbol, Apache Felix, Apache Jena, MongoDB, GATE Jape, OpenNLP NER, Semantic Web/Search, Lexical State Machine, Jsoup/Selenium, Apache POI, Apache Clerezza



Description

The idea of the project is to forecast the demand of goods based on factors from one of the largest retailer in the world. It needed a very large semantic knowledge graph built into it and is able to annotate incoming product pages using this knowledge graph. Categorized 40 million products to 12 super categories, 78 main categories, and 996 subcategories. Used OpenNLP NER to identify terms in the product pages and a custom state machine library and TF-IDF to disambiguate the entities identified for each term. This terms and entities are made into an N-Triple document and stored using Apache Jena into a Triple Store DB for a SPARQL-based search over each item. Semantic search is enabled by Apache Stanbol Content Hub with custom ontology. GATE Jape was used to extract different metadata corresponding to each product in subcategory level.

SafeGuard

URL

<http://msoc.io>

Role

Key Backend Developer

Key Technologies

Dropwizard, Java, OWASP Zap/ Vega Scanner, MongoDB, OpenTSDB, Gnuplot, GCM/APNS Push, Jsoup, Selenium, SockJS, MaxMind GeoIP API.

Description

This project is a secure data communication application for mobile devices in a secure environment. This application will constantly monitor all the network transactions happening through client phone. On the backend, these network events are stored in a time series database backed by OpenTSDB. These events are analyzed for anomaly detection, vulnerability penetration testing (OWASP ZAP), and provide alerts for users. The users can either blacklist or whitelist those websites. Users are also allowed to follow global blacklists issue by admins.

CareerLabs

URL

<https://careerlabs.com>

Role

Backend Developer

Key Technologies

Dropwizard, Java, MongoDB, Stripe/Paypal Payment APIs, iDevAffiliate SDK, MailChimp API, GCM/APNS Push, Jsoup, Selenium, SockJS

Description

This job-search site allows job seekers to filter positions by things like above average work-life balance, great workplace culture, employee satisfaction, employees that share a candidate's political views, and the financial health of a company. CareerLabs uses machine learning and aggregates data from a wide variety of sources, including Freedom of Information Act requests, exclusive surveys, the Bureau of Labor Statistics, the Securities and Exchange Commission, and the Financial Industry Regulatory Authority. CareerLabs offers a basic version of the site for free, but if you want to learn which companies sponsor the most work visas or a business's political alignment, you'll have to pay.

TempStars

URL

<http://tempstars.ca>

Role

Key Backend Developer

Key Technologies

Dropwizard, Java, MongoDB, Stripe/Paypal Payment APIs, GCM/APNS Push, SockJS, Unirest/URLConnection, Managed Pool, Java Mail, iText PDF Writer

Description

TempStars is a premium dental hygiene temp service that uses state-of-the-art mobile technology to quickly, easily and inexpensively match available hygienists with dental offices that need short-term or short-notice hygiene services. TempStars matches available hygienists with dental

offices based on location, rating, and hourly wage. Dentists pay the hygienist her hourly rate directly when the job is done.

Nikaza Beacon AdServer

URL

<http://nikaza.me>

Role

Key Backend Developer

Key Technologies

Dropwizard, Java, MongoDB, OpenTSDB, GnuPlot, RabbitMQ/MQTT, Stripe/Paypal Payment APIs, GCM/APNS Push, SockJS, Java Mail

Description

Nikaza is a Software and Hardware platform for micro location-based contextual engagement and advertising platform based on IoT Bluetooth beacons. The primary goal is to provide ads from participating affiliates to users visiting the beacon area. Tracking and visualization of the effectiveness of ads are available to affiliates through the dashboard.

FoldSchools

URL

<http://cosine.in>

Role

Backend developer, Ideation

Key Technologies

Dropwizard, Java, MongoDB, GCM/APNS Push, SockJS, Java Mail

Description

Foldschools is a school management system and related client mobile apps. It has 4 modules

Student Module: All student activities, complete information of a specific student including personal details, mailing address, last fee payment details, attendance report, etc.

Parent Module: parents can view reports on attendance of their child, performance at studies or in co-curricular activities, etc. They can take a look at the timetable, schedule of the week, reports on tests, exams, etc.

Teachers Module: the creation of schedules, releasing examination details, report submission, forums, etc., are tied with this module. This module can communicate with all other modules.

Admin Module: a way to streamline every management or administration activity at schools. It offers a convenient way to interact or communicate with your school community or others who are directly or indirectly a part of your school.

Eyenvav

URL

<http://www.eyemedvisioncare.com/eyenvav>

Role

Database Designer/Schema Developer

Key Technologies

ER Diagrams, MySQL, Java

Description

A survey based plan chooser which allows the client to navigate through a set of questions. Each question is selected according to responses to the previous question(s). At the end of the survey, the user will be presented with the most relevant eye care plan available from a client according to his/her answers.

StatFuel

URL

<http://statfuel.com>

Role

Backend Developer

Key Technologies

Java, Dropwizard, MongoDB, Stripe/Paypal Payment APIs, GCM/APNS Push, SockJS, Java Mail

Description

StatFuel is the best place to locate any sports statistic within seconds, via search. StatFuel give users the ability to ask about any player, game, stat, or score, and return the precise data point they are looking for. It is the fastest and easiest application for custom sports statistics.

Undisclosed Project

Role

Backend Developer

Key Technologies

Java, Dropwizard, AES, Base64, end-to-end encryption, PostgreSQL

Description

For this project, I had to make a web interface which allows basic user management module. Where users will be able to register, login, logout, forgot the password, reset the password, etc. It used PostgreSQL as database and needed https and end-to-end encryption (AES).

MayDay

Role

Key Backend Developer, Ideation

Key Technologies

Java, Dropwizard, MongoDB, SSE, Lucene, GCM/APNS Push, SockJS

Description

Internal project to make a small app stream audio of our choice to our phones and systems inside our intranet. This app had the ability to create playlists, stream music, shuffle, repeat, etc. The music files are indexed by an Apache Lucene index. Users are allowed to follow other user's playlists.

cTakes Demo

URL

<http://www.auxolabs.in>

Role

Backend Developer

Key Technologies

Java, Dropwizard, MongoDB, SSE, Solr/Lucene, SockJS

Description

cTakes Demo was an app to show the capability of our company to a client. I was responsible for building the backend APIs consumed by the web interface which controls the cTakes module.

MyPeepz Social Network

URL

<http://mypeeps.me>

Role

Key Backend Developer

Key Technologies

Java, Dropwizard, MongoDB, Solr/Lucene, GCM/APNS Push, Java Mail, PBKDF2, OpenFire XMPP, IgniteRealTime Smack, Xuggle, Guice, Gson, Cron4J, MaxMind GeoIP.

Description

MyPeepz is a location-based social discovery network that helps you find people near you that share your background. Whether living or traveling abroad you can find the share and connect with people that come from the same country or region as you or share your culture. Post photos or discuss topics important to you or connect, follow and chat with new friends. Other features include suggested users, photo-sharing (gif/png/jpg/webp), video sharing and streaming (mp4/3gp/mov), GeoJSON, etc.

ECommRecommender

URL

<http://blacktri.com>

Role

ML Developer, Backend Developer

Key Technologies

Java, Dropwizard, MongoDB, Prediction.io, PBKDF2, Machine Learning, Recommendation Engine

Description

ECommRecommender is a configurable e-commerce recommendation engine to identify the performance of the brands under a multi-national conglomerate. It consumes all the transactions



towards the businesses under a management and recommendation will predict the outcome, and most profitable business out of it.

HCare

Role

Key Developer, Designer

Key Technologies

Android, Java, GCM Push, SQLite, Volley, Bluetooth, Adobe Photoshop/Illustrator

Description

H-Care is a client project prototype designed and developed for an undisclosed medical equipment client. This app will work with different medical devices by the company to track patient details. The emergency numbers stored in the app will notify via push notifications and SMS in case of a medical anomaly reported by connected devices.

Urgance 'd Memorizer

Role

Key Developer, Designer

Key Technologies

Android, Java, GCM Push, SQLite, Volley, Reorderable ListView, Adobe Photoshop/Illustrator

Description

Urgance'd Memorizer (Emergency Memorizer) is an app developed for an undisclosed French University for their Safety Course. It was a course material and training app. It also featured a benchmark system to assess the knowledge of students through optional quizzes and etc.

Namaskarangal/Basic Catholic Prayers

Role

Key Developer

Key Technologies

Android, Java



Description

Namaskarangal is a small app to Catholic prayers in the Malayalam language. Where Basic Catholic Prayers translated app in the English language. This app focuses on the essential beliefs and practices of the Catholic Church.

Edusolutions

Role

Key Developer, Designer

Key Technologies

Android, Java, Gestures, Text-to-Speech API, Adobe Photoshop/Illustrator

Description

Edusolutions is a series of educational apps for children. To see, hear, and interact with letters and sounds in words, numbers, and colors. They learn to recognize letters and develop skills that will ensure that the toddler learns the letters sounds much faster.

Solutions Portal

Role

Tester

Key Technologies

JMeter, junit, Gatling

Description

Solutions Portal is a student portal website from Solutions Groups of Institutions; Solutions Portal Mobile Client is an android app which allows registered students to get the services provided in a website to android mobile phones. This is mainly targeted for schools and college students. It has an online public access catalog facility provided for the speedy retrieval of several kinds of documents.



Amy Showcase

Role

Key Developer

Key Technologies

Android, Java Apache HTTP Client

Description

Amy Gold and Diamonds is a client project for showcasing different designs of jewelry from them. It was a privately distributed app for their customers.

